Kingdom of Saudi Arabia

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Beach Scene With Boat Animation

Group members:

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Course Name : Computer Graphics

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May 8 , 2026

Introduction:

Computer graphics are essential for creating interactive visual environments in modern software. This project implements a graphical object-based story using C++ and the Turbo C library, depicting a serene beach scene with a sailboat and a radiant sun. The primary objective is to demonstrate fundamental graphics concepts, including shape rendering and coordinate transformations.

The project is presented in two versions: a **Static Version** with fixed coordinates and a **Dynamic Version** that introduces real-time animation through smooth boat movement. By implementing these versions, we explore the transition from static imagery to graphical animation while ensuring a creative and original visual experience.

Methodology:

The implementation of the "**Beach Scene with Boat Animation**" project followed a modular collaborative approach. The development was divided among four team members, each focusing on a specific component of the graphical environment to ensure a comprehensive and creative scene. The methodology followed these steps:

- **Environment and Background (**Kholod**):**

This phase involved setting up the overall layout of the scene. Using coordinate-based rendering, the sky, beach, and the deep blue sea were implemented. The background was established using setbkcolor and rectangle functions, filled with appropriate colors using floodfill to create a distinct horizon and shoreline.

- **Celestial Objects and Atmosphere (**Marya**):**

This part focused on drawing the sun and clouds to enhance the sky's visual appeal. The sun was rendered using the **Midpoint Circle Algorithm** for precise circle generation. For the clouds, a series of overlapping fillellipse functions were used to create a smooth and organic appearance.

- **The Sailboat Design (**Fatimah**):**

The core object of the scene, the sailboat, was constructed using a combination of complex polygons and lines. The **DDA (Digital Differential Analyzer) Algorithm** was implemented to draw the structural lines of the boat and the mast with high precision. fillpoly was utilized to render the boat's hull and the two-toned sails, adding a sense of depth to the 2D object.

- **Dynamic Animation and Motion Control (**Rama**):**

The project was finalized by transforming the static scene into a **Dynamic Version**. This was achieved by implementing a continuous while loop that clears the screen and redraws the objects with updated horizontal coordinates. Specifically, the sailboat was programmed to move across the sea from left to right, while the clouds and sun remain as static elements of the background.

Algorithm:

The project relies on two fundamental computer graphics algorithms to render the scene's structural elements with mathematical precision:

A. DDA (Digital Differential Analyzer) Line Algorithm

The DDA algorithm was implemented to draw the straight structural lines of the sailboat and its mast. It follows an incremental approach based on the following steps:

1. Calculate Differences: Determine the horizontal and vertical distances between the start point (x_1, y_1) and end point (x_2, y_2) using dx = x_2 - x_1 and dy = y_2 - y_1.

2. Determine Steps: The number of steps required is the maximum absolute value of either dx or dy.

3. Calculate Increments: The change required per step for both coordinates is calculated as xInc = dx / steps and yInc = dy / steps.

4. Pixel Plotting: Starting from the initial point, the coordinates are incrementally updated and plotted using the putpixel function, with values rounded to the nearest integer to ensure a continuous line.

B. Midpoint Circle Algorithm

The Midpoint Circle Algorithm was utilized to render the sun. This algorithm is highly efficient as it uses integer-based decision parameters to determine the optimal pixel path:

1. Initialization: The process begins at the point (0, r), where r is the circle's radius.

2. Decision Parameter: An initial decision parameter d = 1 - r is calculated to evaluate the relative position of pixels.

3. Eight-way Symmetry: To optimize performance, the algorithm calculates points for only one-eighth of the circle.

These points are then reflected across all eight octants using symmetry.

4. Coordinate Update: In each step, x is incremented, and based on the value of d, the algorithm decides whether to maintain or decrement the y coordinate, ensuring a smooth circular boundary.

Discussion:

During the development of this project, we faced several challenges. We also faced challenges in designing the boat using coordinates and ensuring that all parts were aligned correctly.

Another challenge was implementing smooth animation. This was achieved by using a loop, continuously updating the position of the boat, and redrawing the scene.

Dividing the project among team members helped us organize the work efficiently. Each member focused on a specific part, which improved the overall quality of the project.

Conclusion:

In this project, we successfully created a 2D beach scene that includes both static and dynamic elements. We learned how to use basic computer graphics functions to draw shapes and build .complex objects such as a boat

We also understood how animation works by updating object positions over time. This project improved our programming .skills in C++ and enhanced our ability to work as a team

Overall, the project helped us apply theoretical concepts of .computer graphics in a practical way